

Laboratory work #2

Academic year 2023/2024, autumn semester

Course: **Algorithmization and Programming of Solutions (DIP107)**, Part I

Individual work

Branching and looping

Create a program that processes data inputted by a user according to the task variant and draw a corresponding flowchart. (2p)

A list of steps:

1. The user of the program enters the value "x". (1p)
2. The program computes two intermediate values "a" and "b", and outputs these values on the screen (3p)
3. The program computes and outputs the final result "y" using the first or the second expression, depending on the given condition: (2p)

$$y = \begin{cases} \text{the first expression, if the condition } C \text{ is TRUE} \\ \text{the second expression, if the condition } C \text{ is FALSE} \end{cases}$$

Additional requirements:

1. The program has to output text messages, if computation errors occur (for example, dividing by 0) (1p)
2. The program has to provide an opportunity to repeat all action as many times as a user wish. (1p)
3. A programming language – C.

To get your task variant, take the last 2 digits of your student ID and subtract 20 until you get a number between 1 and 20
Example: last digits of ID is "77".

$77 - 20 = 57 - 20 = 37 - 20 = 17$. Your task variant is Nr.17

Task variants

Number	a	b	1 st expression	2 nd expression	C
1.	$\frac{5x}{2x^3}$	$ 2x - 3x^2 $	$a + b(a + b)$	$a^2 - b^2$	$a^2 > b^3$
2.	$5x + 33$	$2x - 7x^3$	$a + b + 2ab$	$a(b - 1)$	$\frac{100}{a} \geq b$
3.	$x^2 + 10$	$2 x - 5 $	$b\left(\frac{a+b}{2} + \frac{2}{a-b}\right)$	$\frac{a}{2} + ab$	$a \neq 18$
4.	$x^2 - x$	$\pi x + x^2$	$a^2 - b + ab$	$\frac{1}{4}\left(\frac{2}{a} - \frac{b}{10}\right)$	$a \geq b$
5.	$5x + 3x^3$	$x^2 + 2x$	$(a - b)^2$	$-\frac{1}{2}\left(\frac{a}{4} - \frac{2}{a+b}\right)$	$a - b > 0$
6.	$3x^2 + 5 x $	$\frac{9}{2+3x}$	$a - 2a^2 + b$	$\frac{2b}{5} - \frac{2a^2}{7}$	$b \leq 10$
7.	$x^3 + x $	$7x + 5$	$a(2a - b^2)$	$2a - \frac{2}{b}$	$a \leq b$
8.	$\frac{x}{\pi - x} - 2$	$x^2 + x - 8$	$2(b - 1)a$	$2a + b^3$	$a + b \leq 10$

9.	$\frac{1}{12x^2 + 7x - 5}$	$ x^2 - x^3 $	$4a^2 + 2b^4$	$\frac{b}{2} + \frac{a}{2}(a + b)$	$a \leq 10$
10.	$\frac{x^2 + 13}{x}$	$(x + 1)^2$	$(a + b) - \frac{b}{2}$	$1 + a^3b^2$	$a + 3 \geq b$
11.	$\frac{x^2 + 3}{18x - 1}$	$x - 2 + (1 + x)^2$	$3a^3 + 3b^3$	$(a - 1)(b - 1)$	$a \geq b + 10$
12.	$3x - 16$	$1 + \sqrt{x - 8}$	$a^2 + ab + b^2$	$\frac{a}{2}(b + ab)$	$a \geq 25$
13.	$4x^3 - 1 - \sqrt{x}$	$x + x^4 - x^5 $	$6a(a^2 + b)$	$\frac{a + b}{10}$	$a \leq b$
14.	$\sqrt{x^3 - 10x}$	$x^3 + x + 5$	$\frac{a^2 + b^2}{4}$	$a(b - 1)(a - b)$	$a^3 \neq 0$
15.	$\frac{x + 7}{4}$	$\frac{ x }{1 + x}$	$3ab^2 - 2b$	$b + \frac{a^3}{10}$	$a^2 > b$
16.	$\frac{x^2 + x}{2 x - 10 }$	$x^2 + 16x - 2$	$a^3 - b^3$	$\frac{ab}{2} + a$	$a + 10 \geq b$
17.	$\sqrt{10x - x^2}$	$\frac{10x - 4}{5x - 3}$	$\frac{a + b^2}{5}$	$\frac{a^2}{2} + \frac{b}{10}$	$a \geq b$
18.	$\frac{x^2 - 7x + 10}{x^2 - 8x + 12}$	$3x + 4$	$\frac{a^3}{3} - 2b - \frac{b^2}{2}$	$\frac{a + b}{2} - \frac{a - b}{4}$	$a \geq 9$
19.	$x^2 - 2x$	$\sqrt{x + x^2 - 10}$	$\frac{a}{2} + \frac{a + b}{2}$	$5 \times ((a + b) + 3)$	$a > 35$
20.	$\frac{1}{x} - \frac{x^3}{3} + \frac{x^2}{5}$	$x - 14$	$\frac{a^3}{3} + \frac{b^4}{4}$	$\frac{a + b}{2} + (a - b)$	$a + b \geq 6$